

OUTPUTS

QUESTION 1: Doctor shift scheduling using backtracking search

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\monik\AppData\Local\Programs\Python\Python312\output ai 1.py
Final Valid Schedule:
{'D1': 'Morning', 'D2': 'Afternoon', 'D3': 'Night'}
>>>
```

QUESTION 2: BFS

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\monik\AppData\Local\Programs\Python\Python312\output ai 1.py
Optimal Path: [(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]
Cost: 8
>>>
```

QUESTION 3: Autonomous rescue owner using uniform cost search

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\monik\AppData\Local\Programs\Python\Python312\output ai 1.py
Least-Cost Path: [(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]
Total Cost: 8
>>>
```